

MARTC: Multi-Agent Retrieval with Task-State Control

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Abstract—Large language models can generate fluent responses, but specialized assistance like tutoring requires grounding in external evidence, visible user or task state, and explicit answer criteria. Classical retrieval-augmented generation (RAG) usually treats grounding as a fixed retrieve-then-read step in which a query is matched against a corpus and the generator reads the returned passages. This control flow is rigid when the same surface question requires different evidence, response structure, or evaluation criteria across situations. We propose MARTC (Multi-Agent Retrieval with Task-State Control), a multi-agent retrieval framework that treats RAG as structured prompt construction rather than query-only passage lookup. A supervisory router decides which context requirement should dominate the prompt (planning, criteria, adaptation, or evidence) and whether retrieval should run; LLM-backed planning, state diagnosis, criteria analysis, and conditional retrieval prepare structured context blocks; and all intermediate outputs are reassembled into one augmented prompt for a fixed generator followed by one critique/revision pass. We instantiate and evaluate MARTC in educational tutoring, a setting where evidence, learner-visible state, and answer criteria must be coordinated. Controlled validation on two educational benchmarks under matched backbone, corpus, reranker, and response cap shows that MARTC improves quality while reducing token use, with additional latency and model-call reductions in some settings.

I. Introduction

Large language models (LLMs) are increasingly used as specialized assistants because they can generate explanations, diagnose errors, and provide feedback. However, useful assistance cannot rely only on parametric knowledge. Knowledge-grounded dialogue has long shown the value of external evidence [1]; many assistance settings require grounding in curated resources, user-visible context, task history, and evaluation criteria to avoid fluent but misaligned guidance. Educational tutoring is a concrete and demanding instance of this problem: a tutor must coordinate course materials, worked examples, rubrics, learner-visible context, and dialogue history [2], [3].

Classical retrieval-augmented generation (RAG) has therefore become the default grounding mechanism [4]. In classical RAG, the query is embedded, the top- k most similar passages are retrieved, and the LLM generates a response conditioned on those passages. Thus, relevance depends heavily on retrieval quality and query representation [5].

Specialized assistance, e.g., tutoring, is more complex than factual question answering. The appropriate response may depend on user background, current goal, task history, prior confusion, and evaluation criteria. We call these visible attributes the user’s personalized learning state. A response must therefore be grounded not only in topically relevant evidence, but also in the user’s state and task-specific requirements.

Classical RAG can include explicit state text when supplied, but appending user state, dialogue, criteria, and task constraints forces one query embedding to compress topic, level of detail, intent, and response requirements. As constraints grow, search intent can blur. The issue is therefore how a response system should organize constraints before retrieval and generation.

Therefore, we introduce MARTC (Multi-Agent Retrieval with Task-State Control), a multi-agent RAG framework for LLM-based tutoring. The core idea is to move beyond query-only retrieval by organizing the tutoring context before retrieval and generation. MARTC uses a supervisory regime design to determine what the response should prioritize. Each regime then activates the relevant specialized agents to construct the context needed for that type of tutoring response. In this way, MARTC avoids forcing all tutoring needs into a single retrieval query and instead separates search intent, learner state, answer criteria, and instructional evidence into structured context blocks. Retrieval is guided by a clearer purpose, while learner-state and criteria information remain available throughout generation and post-generation critique.

Contributions. (i) We propose MARTC, a multi-agent retrieval framework with routing, specialist context construction, bounded tools, conditional retrieval, generation, and critique. (ii) We formalize transparent training-free controls over task state, answer criteria, and external evidence. (iii) We instantiate the framework for educational tutoring and validate it against classical RAG under paired infrastructure, with an additional agentic coordination snapshot for comparison.

II. Related Work

RAG controllers. Classical RAG fixes retrieval as the first controller: the query is matched to passages, and generation conditions on returned evidence [4]. This ordering is brittle when the same surface query should be constrained by visible state, answer criteria, or response form before evidence is selected. Recent variants improve

specific control points. Self-RAG learns reflection tokens for retrieval and critique, so it cannot be adopted without training or instruction tuning the backbone [6]. CRAG evaluates already-retrieved documents and optionally expands to web search, so it corrects retrieval after the initial query rather than deciding what constraints should shape that query [7]. Adaptive-RAG trains a smaller model to classify question complexity and choose among QA strategies, which controls reasoning depth but not task state, criteria, or prompt assembly [8]. TARG is training-free and efficient, but its decision is only a retrieve-or-skip gate computed from a no-context draft; it does not build structured context before retrieval [9]. RAG diagnostics separate retrieval, ranking, and generation errors, but they diagnose failures rather than provide a controller that routes, budgets, retrieves, and assembles context [10]. The remaining gap is not another retriever; it is a controller that organizes constraints before retrieval.

Agentic retrieval. Agentic RAG and multi-agent LLM systems address the flexibility missing from fixed pipelines, but their control loops are usually open-ended. ReAct interleaves reasoning with tool actions, while AutoGen, CAMEL, and MetaGPT coordinate conversable or role-playing agents for broad task execution [11]–[14]. MA-RAG is closer because it decomposes ambiguous and multi-hop QA into planner, step-definition, extraction, and QA agents [15]. Its strength is iterative reasoning for difficult questions; its limitation for controlled RAG is that agents remain part of an answer-solving chain rather than a bounded prompt-construction contract.

Educational agents and evaluation. Educational systems such as AgentTutor assess learners, maintain memory, and adapt teaching strategies across multi-turn instruction [3]; evaluation work measures whether a produced tutor response is pedagogically useful [2], [16]–[19]. These works motivate our validation setting because they show that visible state and answer criteria matter. However, complete tutoring policies entangle retrieval quality with memory, dialogue policy, and teaching strategy. They therefore do not isolate the question MARTC targets: whether context construction itself can be made more flexible and efficient under a fixed generator.

III. Methodology

As stated earlier, educational tutoring requires more than answering a question correctly. A tutor often needs to decide what the learner is asking, what concept or misconception is involved, what level of explanation is appropriate, what criteria should be satisfied, and whether external instructional evidence is needed. These requirements make tutoring a context-construction problem before it is a generation problem. Simply appending these constraints to a query in classical RAG can overload the query representation, blur search intent, and retrieve evidence that is topically relevant but not pedagogically appropriate. Although adaptive RAG

methods start to address this issue, they tend to target one control problem at a time. This motivates a unified multi-agent framework that coordinates learner state, instructional goals, answer criteria, and evidence needs before generation. MARTC is then designed to address this need, which treats RAG as coordinated context construction rather than query-only passage lookup. As shown in Fig. XX, MARTC consists of three main components: input normalization, supervisor routing, and specialist agent and coordination.

A. Input Entry and Inference Caps

For one sample, MARTC first standardizes all visible fields into

$$x = (q, o, c, h, r, v). \quad (1)$$

where q is the question or task; o answer options; c inline context; h recent interaction history; r explicit criteria; and v images. Empty fields are allowed, so the same representation covers factual questions, assessment, feedback, and planning.

The only user-facing output is answer y . Internally, MARTC keeps the route choice, retrieval decision, evidence bundle D , and tool observations T . Planner output, state summary, criteria summary, T , and D become prompt blocks. Caps keep inference bounded: K for planner/retrieval queries, Q for chunks/tool calls, L for evidence length, and N for response tokens.

The core design choice is to avoid asking one retrieval query to carry every constraint. MARTC instead fills three structured slots. The planner writes $p = (a, P)$, where a is a short answer strategy and P is a capped list of search queries that tell the retriever what to look for. The state diagnoser writes state summary ℓ , inferred only from q and h . The criteria analyst writes criteria summary u , preserving explicit criteria r or supplying a short correctness/clarity/evidence/actionability checklist. This separates search, task state, and answer requirements [3], [20].

B. Supervisor Routing

The supervisory router is responsible for mapping each input sample to a pedagogical regime and activating the corresponding specialist agents. Given an input sample, the router examines the visible question, optional context, dialogue history, learner-state information, rubric or criteria text, and task metadata. It then estimates four types of context need: planning, criteria feedback, learner-state adaptation, and evidence grounding.

These needs correspond to four regimes. Planning (PL) is selected when the response requires an ordered instructional structure, such as lesson planning, step sequencing, or exercise design. Criteria Feedback (CF) is selected when the response must be judged against explicit requirements, such as grading rubrics, scoring rules, or required feedback fields. Adaptive Response (AR) is selected when the same content should be explained differently depending on the learner’s visible

state, such as grade level, prior confusion, weak points, or preferred explanation style. Instructional Material Grounding (EG) is selected when the response should be supported by external instructional materials, such as textbook chapters, reference documents, or worked examples.

To make this routing decision transparent and training-free, MARTC uses cue-based scores rather than a learned classifier. The route first constructs a routing string $b(x)$ by concatenating the textual fields that are visible before generation:

$$b(x) = \text{concat}(q, c, r, h), \quad (2)$$

where $\text{concat}(\cdot)$ denotes the normalization and concatenation process used to form a single routing string. Images remain available to the downstream components, but they are not converted into the textual routing score.

The router then evaluates cue families $j \in \{\text{ev}, \text{cr}, \text{pl}, \text{ad}\}$, corresponding to instructional material grounding, criteria feedback, planning, and adaptation. Each family has keyword set W_j ; let $M_j(x)$ be the number of cues from W_j in $b(x)$. The normalized cue score is

$$s_j(x) = \frac{M_j(x)}{|W_j|}, \quad (3)$$

where $|W_j|$ is the size of the j th cue-family.

The routing thresholds are introduced to prevent weak or incidental cue matches from changing the system behavior. Because the router is training-free, these thresholds serve as transparent control parameters rather than learned decision boundaries. The general threshold τ_{mid} requires repeated cues before a context need is activated. The planning threshold τ_{plan} is set more strictly because selecting the planning regime changes the response structure, not only its content. The retry threshold τ_2 is reserved for fallback retrieval and is therefore used only when the first retrieval attempt fails but the evidence need remains strong.

Based on these cues and thresholds, the supervisor assigns the pedagogical regime as follows:

$$R(x) = \begin{cases} \text{PL}, & s_{\text{pl}} \geq s_{\text{cr}}, s_{\text{pl}} \geq s_{\text{ad}}, s_{\text{pl}} \geq \tau_{\text{plan}}, \\ \text{CF}, & s_{\text{cr}} \geq s_{\text{ad}}, s_{\text{cr}} \geq \tau_{\text{mid}}, \\ \text{AR}, & s_{\text{ad}} \geq \tau_{\text{mid}}, \\ \text{EG}, & \text{otherwise.} \end{cases} \quad (4)$$

where $R(x)$ is the regime-selection function.

In addition to selecting the pedagogical regime, the supervisor also makes a binary retrieval decision, denoted by $G(x)$ to determine whether external evidence should be retrieved and added to the final prompt. needed [6], [8], [9]:

$$G(x) = \begin{cases} 1, & s_{\text{ev}} \geq \tau_{\text{mid}} \text{ or } R(x) = \text{EG}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Thus, retrieval is activated if the evidence cue score is high enough, or if the selected regime is Instructional Grounding. Otherwise, the system does not retrieve external passages.

C. Specialist Agents and Coordination

After the supervisor assigns a regime, MARTC dispatches the input to the relevant specialist components in two stages. In the pre-generation stage, the planner, criteria agent, and diagnoser construct the tutoring brief. These agents do not engage in open-ended conversation with one another; instead, each produces a structured record that is merged by the coordinator. The planner identifies the instructional goal and response strategy. When instructional-material grounding is needed, it also generates scoped search queries that specify what external materials should be retrieved. These queries are later passed to the retriever, which is activated separately by the route or retrieval gate described in the previous subsection. The criteria agent summarizes explicit rubrics, grading rules, required output fields, or default answer-quality criteria. The diagnoser summarizes the learner's visible state, including level, goals, possible misconceptions, prior confusion, or preferred explanation style.

In the post-generation stage, the critic agent operates after the tutor generates an initial response. It is not part of the brief-construction stage. Instead, it uses the tutoring brief, retrieved evidence, and draft response to perform one bounded revision pass, checking for appropriate use of evidence, coverage of criteria, and pedagogical alignment with the learner state. If issues are detected, the critic revises the response once; otherwise, the draft is returned as the final answer.

The key point is that each regime changes how the tutoring brief is constructed. MARTC always begins with a default task prompt containing the original input and general tutoring requirements. When a regime is activated, MARTC adds regime-specific context blocks to the tutoring brief.

In PL, the planner provides the main organizing structure, such as the instructional goal, step sequence, or response strategy. In CF, the criteria agent defines the evaluation standard, and the critic later checks whether the generated response satisfies it. In AR, the diagnoser supplies the learner-state summary that shapes tone, depth, scaffolding, and feedback. In EG, the planner formulates scoped search intent, and the retriever supplies instructional-material evidence when the retrieval gate is open. The final tutor prompt is assembled from the default task prompt plus the selected regime, plan, learner-state summary, criteria summary, retrieved evidence when available, and task instructions.

This agent-regime design avoids forcing all tutoring needs into a single retrieval query. Instead of embedding one long query containing the topic, learner state, rubric, response goal, and evidence need, MARTC decomposes these requirements into structured prompt blocks. Retrieval is guided by clearer search intent, while learner-state and criteria information remain available as separate context for generation and post-generation

critique.

Algorithm 1: MARTC Inference Process

Require: entry $x = (q, o, c, h, r, v)$, caps K, Q, L, N , index $\mathcal{I}$, tools $\mathcal{U}$, thresholds $(\tau_{\text{mid}}, \tau_{\text{plan}}, \tau_2)$.
 Return: answer y and internal fields (R, G, p, ℓ, u, T, D) .

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1:  $b \leftarrow \text{concat}(q, c, r, h)$ ;  $(s_{\text{ev}}, s_{\text{cr}}, s_{\text{pl}}, s_{\text{ad}}) \leftarrow \text{CueScores}(b)$ 
2:  $R \leftarrow R(x)$ ;  $G \leftarrow G(x)$ 
3: parallel do
4:    $p = (a, P)$ ;  $T_p \leftarrow \text{Planner}(x, R, G, \mathcal{U})$ 
5:    $\ell, T_\ell \leftarrow \text{State}(q, h, \mathcal{U})$ 
6:   if  $r \neq \emptyset$  or  $R = \text{CF}$  then  $u, T_u \leftarrow \text{Criteria}(x, R, \mathcal{U})$ 
7:   else  $u \leftarrow \text{DefaultCriteria}()$ ;  $T_u \leftarrow \emptyset$ 
8: end parallel
9:  $P \leftarrow \text{cap}(P, K)$ ;  $p \leftarrow (a, P)$ ;  $T \leftarrow \text{cap}(T_p \cup T_\ell \cup T_u, Q)$ ;  $D \leftarrow \emptyset$ 
10: if  $G = 1$  then
11:    $D \leftarrow \text{Pack}(\text{Rerank}(\text{Dedup}(\text{Search}(\mathcal{I}, P))), q, Q, L)$ 
12: end if
13:  $D_0 \leftarrow D$ 
14:  $y_0 \leftarrow A_{\text{gen}}(\text{Prompt}(q, o, h, p, \ell, u, T, D, c), N)$ 
15: if  $G_2(x) = 1$  then
16:    $D \leftarrow \text{Pack}(\text{Rerank}(\text{Dedup}(\text{Search}(\mathcal{I}, [q]))), q, Q, L)$ 
17:    $y_0 \leftarrow A_{\text{gen}}(\text{Prompt}(q, o, h, p, \ell, u, T, D, c), N)$ 
18: end if
19:  $y \leftarrow \text{CriticOnce}(y_0, q, h, p, \ell, u, D, r)$ 
20: return  $y, (R, G, p, \ell, u, T, D)$ 

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IV. Validation Experiments

We validate the framework on two public educational benchmarks used as tutoring-focused stress tests. TutorEval [2] pairs science tutoring questions with long STEM chapters and teaching key points; EduBench [17] covers diverse educational scenarios with prompts, meta-data, and reference model predictions that our adapter converts into score and rubric supervision.

The primary validation compares context-assembly methods under matched infrastructure, with a Qwen3.5-27B-FP8 replication checking backbone sensitivity. Classical RAG retrieves first from the raw standardized prompt; visible learner profile, student answer, dialogue, and criteria text remain available to its generator prompt, so the baseline is not deprived of task context. Its limitation is controller order: retrieval happens before specialist summaries can shape search. The 4B table also reports the previous multi-agent non-retrieval run as an Agentic baseline: it uses specialist coordination without the conditional evidence branch, so it tests whether coordination alone explains the gains. MARTC keeps the same specialists and retriever, then adds supervisory routing, conditional retrieval, and retry control. We report quality and cost together because only context construction changes.

A. Implementation Details

All controllers use the same deployed Python framework, corpus, hybrid text index, reranker, generator, critic, cache policy, and metric code; only context assembly differs. The planner, state diagnoser, criteria module, generator, and critic use the same configured backbone; deterministic fallbacks run only when an LLM response is unavailable or malformed. MARTC enables bounded local tool calls for specialist observations and corpus search, with no network, file mutation, or open-ended shell access. LLM endpoints are served through

TABLE I

Quality and resource summary with Agentic and 27B comparison.

Data	Metric	RAG	Agentic	MARTC	MARTC-27B	Δ
TutorEval	Token-F1	0.112	0.090	0.115	0.272	+0.004*
	Tutor hit	0.938	0.932	0.957	0.918	+0.019**
	Rubric	0.919	0.907	0.944	0.891	+0.026**
	Latency (s)	51.90	51.05	49.11	40.65	-2.79
	Tokens	3975	5959	3490	6730	-484***
EduBench	EB12D	0.367	0.398	0.398	0.425	+0.031***
	Rubric	0.705	0.891	0.891	0.769	+0.186***
	EduAlign	0.166	0.130	0.126	0.526	-0.040**
	Latency (s)	19.83	23.73	13.90	32.09	-5.92***
	Tokens	4020	5224	2226	3717	-1794***

Agentic is the prior multi-agent non-retrieval snapshot. MARTC-27B shows the 27B replication for our framework only; the full paired 27B RAG comparison appears in Table II. Bold marks the best displayed value after rounding within each row. Δ and significance compare paired 4B MARTC-RAG only: *** $p < 10^{-4}$, ** $p < 0.01$, * $p < 0.05$. Higher is better for quality; lower is better for latency and tokens. Runtime is mean wall-clock seconds/sample on the experiment server.

SLang, whose runtime is designed for structured multi-call LLM programs and KV-cache reuse [21]. We also enable exact model-response caching for every controller because repeated prompts, fixed criteria, and repeated served requests are normal in deployed systems; a hit requires an identical full payload and is marked in the result artifacts. Therefore latency is cache-aware served-system wall time, not a cold-start model benchmark. Runs use Qwen3.5-4B and Qwen3.5-27B-FP8 [22], reasoning budget 512, generation temperature 0.15, specialist JSON temperature 0.0, and response cap $N=900$ on benwulab-Lambda-Vector with two NVIDIA RTX 6000 Ada GPUs. We set $K=3$, $Q=4$, $L=6000$, and router thresholds $(\tau_{\text{mid}}, \tau_{\text{plan}}, \tau_2) = (0.35, 0.42, 0.45)$.

Evaluation is dataset-aware, and all quality scores are automatic proxies rather than official benchmark or human grades. Inspired by RAG evaluation work [10], we separate answer overlap, criteria fulfillment, grounding, and resource cost. Token-F1 balances reference-token coverage against extra generated tokens; rubric coverage checks whether benchmark criteria are expressed with exact or content-bearing partial matches; latency is served wall-clock seconds per sample under the shared cache policy; and tokens count prompt plus completion tokens.

For EduBench, EB12D averages 12 normalized checks over scenario adaptation, factual/reasoning behavior, and pedagogical application, while EduAlign only measures extracted-score agreement with the benchmark consensus. For TutorEval, tutor-hit measures expert teaching-point coverage with full, partial, or zero credit.

B. Framework Validation

Table I reports paired 4B comparisons with bootstrap CIs and permutation p-values, plus the previous Agentic snapshot for reference and a 27B MARTC replication column. Table II reports the full 27B paired RAG-MARTC comparison. The Agentic column is not used for significance testing because it comes from the earlier completed multi-agent non-retrieval run.

TABLE II
27B quality and resource summary.

Data	Metric	RAG	Ours	Δ
TutorEval	Token-F1	0.264	0.272	+0.008*
	Tutor hit	0.897	0.918	+0.021*
	Rubric	0.861	0.891	+0.030*
	Latency (s)	41.47	40.65	-0.82**
	Tokens	6681	6730	+49
EduBench	EB12D	0.355	0.425	+0.070***
	Rubric	0.634	0.769	+0.135***
	EduAlign	0.541	0.526	-0.015
	Latency (s)	41.91	32.09	-9.81***
	Tokens	6105	3717	-2388***

TutorEval uses $n=400$ paired samples; EduBench uses $n=328$ unique paired rows after duplicate public IDs in the 400-example slice are collapsed. *** $p<10^{-4}$, ** $p<0.01$, * $p<0.05$. Higher is better for quality; lower is better for latency and tokens. Runtime is mean wall-clock seconds/sample on the experiment server.

C. Trade-offs

For TutorEval, the 27B paired CIs are positive for Token-F1, tutor-hit coverage, rubric coverage, and latency. For EduBench, EB12D and rubric coverage improve, latency and token use decrease, and the small classical-RAG EduAlign edge is not statistically reliable.

MARTC can use more modules than classical RAG, so resource cost must be reported. Scoped planner queries and the retrieval gate reduce token use in 4B and still reduce latency in 27B, but 27B TutorEval shows that better coverage does not always mean fewer tokens. Thus MARTC decides what grounding a request needs; it is not a universal retrieval replacement.

V. Limitations

The study validates a framework rather than a deployable assistant: evaluation is automatic and English-only, and user or task state is inferred from visible prompt/dialogue fields. Deployment requires privacy safeguards, bias checks, data governance, human oversight, and domain-specific outcome studies.

VI. Conclusion

MARTC reframes grounding as supervised context construction rather than query-only passage lookup. With shared infrastructure, the educational tutoring instantiation improves fixed-generator quality and efficiency over classical RAG by coordinating visible task state, answer criteria, and conditional evidence before generation. Future work should test stronger controller baselines, other instructional settings and non-educational domains, human and multi-turn outcomes, learned router calibration, profile-aware reranking, and privacy-preserving state inference.

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